

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Summer Examination – 2023 Course: B. Tech. Branch : Mechanical Engineering Semester: VI Subject Code & Name: BTMC601 & Manufacturing Processes-II Max Marks: 60 Date: Duration: 3 Hrs.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		Level	Marks
Q. 1	Solve Any Two of the following.		12
A)	Define abrasive machining and explain its importance. Draw a schematic diagram of grinding process.	<i>Remember</i>	6
B)	What are the super-abrasives? Which super-abrasive is recommended for grinding of steels and why?	<i>Understand</i>	6
C)	Define G-ratio in grinding operation and explain why does it vary widely.	<i>Analyse</i>	6
Q.2	Solve Any Two of the following.		12
A)	What is mechanics of metal cutting? What are its objectives? Draw a schematic diagram of metal cutting showing the workpiece, cutting tool, and chip.	<i>Remember</i>	6
B)	What is rake angle? Explain positive rake angle and negative rake angle with the help of a schematic diagrams. Why is clearance angle provided on the cutting tool?	<i>Understand</i>	6
C)	Define specific cutting energy. What is its importance? What are the parameters affecting it?	<i>Remember</i>	6
Q. 3	Solve Any Two of the following.		12
A)	Explain the principle of measurement of cutting temperature with radiation methods. What is the principle of correlation of cutting temperature with hardness and microstructure changes in the cutting tool material?	<i>Remember</i>	6
B)	Explain the basic requirements of cutting tool materials. What are the major classes of cutting tool materials?	<i>Understand</i>	6
C)	Define machinability and discuss the factors affecting the machinability of metals.	<i>Remember</i>	6

Q.4	Solve <i>Any Two</i> of the following.		12
A)	Define atomization method of producing metal powders. Explain gas atomization and water atomization with the help of schematic diagrams.	<i>Remember</i>	6
B)	Discuss the purposes of blending of metal powders. What are the precautions to be taken during blending of metal powders?	<i>Understand</i>	6
C)	Discuss the sintering mechanisms with the help of schematic diagrams.	<i>Remember</i>	6
Q. 5	Solve <i>Any Two</i> of the following.		12
A)	Describe the processes for producing polymer sheets and films.	<i>Remember</i>	6
B)	Explain injection molding process with the help of a schematic diagram.	<i>Understand</i>	6
C)	Discuss thermoforming process with the help of a suitable diagram.	<i>Understand</i>	6
	*** End ***		

The grid and the borders of the table will be hidden before final printing.